

Evaluation of Monocular and Binocular Contrast Sensitivity on Virtual Reality Head-Mounted Displays

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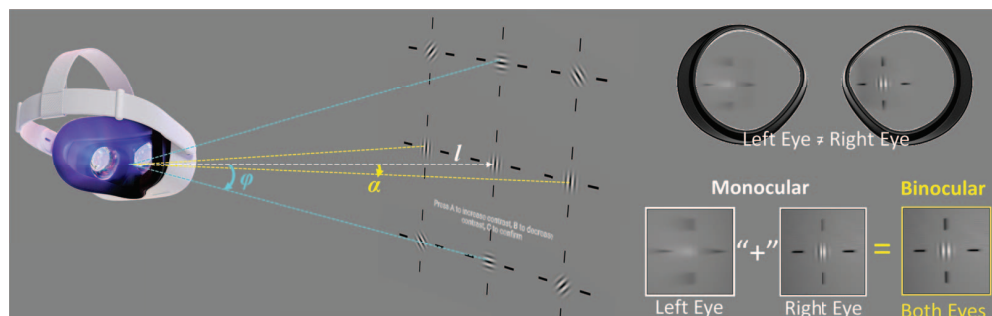


Figure 1: Schematic illustration of the monocular and binocular contrast sensitivity measurements on VR HMDs. Spatial variation of monocular image quality on VR HMDs results in unpaired visualizations on each eye piece.

ABSTRACT

Virtual reality (VR) creates an immersive experience by rendering a pair of graphical views on a head-mounted display (HMD). However, spatial variation of image quality on VR HMDs can potentially lead to nonuniform monocular views and binocular disparity. In this study, we begin to bridge the gap between monocular and binocular image quality evaluation by developing a WebXR test platform to perform human observer experiments. Specifically, monocular and binocular contrast sensitivity functions (CSFs) are obtained using varied interpupillary distance (IPD) conditions. The result demonstrates that monocular image quality degrades when gazing at the periphery of display field of view (FOV) especially for mismatched IPD settings. Binocular contrast perception is dominated by the monocular view on a VR HMD with superior image quality measured by the contrast.

1 INTRODUCTION

Image quality on a virtual reality (VR) head-mounted display (HMD) experienced by the user is dependent on the fusion of images in both eyes. However, spatial variation of image quality [6] and geometric distortion [4] between HMD eyepieces can lead to interocular disparity in image quality and potential binocular rivalry [1]. As illustrated in Fig. 1, optical limitations in VR HMDs can result in monocular spatial nonuniformity and a substantial difference in image quality between left and right eyes. Binocular disparity can be more pronounced if the interpupillary distance (IPD) of the user does not match the mechanical IPD setting of the HMD [6]. Current evaluation of image quality on VR HMDs is generally based on monocular optical bench measurements with test methods established standard documents [2, 3]. Specific requirements on the light measuring devices and bench setup have been recommended in the standards to emulate the eye anatomy and rotation mechanism to investigate gaze-dependent monocular image quality on VR HMDs. However, it is uncertain whether these measurements adequately

capture the binocular image quality experienced by users in MXR applications. In this study, we compare the monocular and binocular contrast sensitivity using human observers. Specifically, we developed a WebXR-based test platform to render test targets with varied contrast (modulation) on the evaluated VR HMD and performed human observer experiments to obtain the monocular and binocular contrast sensitivity using various IPD configurations.

2 METHODS

2.1 Head-mounted display

A Meta Quest 2 HMD was used for human observer experiments to measure the monocular and binocular contrast sensitivity. The Quest 2 offers three IPD settings, i.e., 58 mm (small), 63 mm (nominal), and 68 mm (large), providing users with options to customize the HMD IPD for optimal viewing comfort and visual performance. The pixel resolution of the display backplane is 1832 (horizontal) $\times$ 1920 (vertical) pixels per eye, refreshing at 90 Hz. A pair of Fresnel lenses magnifies the virtual images yielding a FOV of about 90°.

2.2 WebXR target detection platform

We developed a WebXR platform to perform human observer experiments by determining the contrast sensitivity defined as the reciprocal of threshold contrast that displayed a Gabor stimuli for detection. The Gabor target is a 2D sinusoidal pattern with predefined contrast and spatial frequency. The target was placed at 9 locations covering a $18 \times 18^\circ$ field of view. At each location, the orientation of the target was modified such that the sinusoidal contrast pattern is always circular (or perpendicular to the radial direction) with respect to the target location. It has been shown that the circular orientation of the pattern is suitable to capture the contrast degradation by optical aberration [6].

2.3 Human observer experiments

The FDA IRB reviewed and approved the study protocol (IRB 2022-CDRH-038) and all human subject studies were conducted under the approved protocol. We recruited 9 human subjects (with ages between 21 and 47 years, average age 27.7, three females, six males) with normal or corrected to normal vision by prescription glasses to participate in the human observer experiments. The IPD of the subjects was recorded. Human observer experiments were conducted

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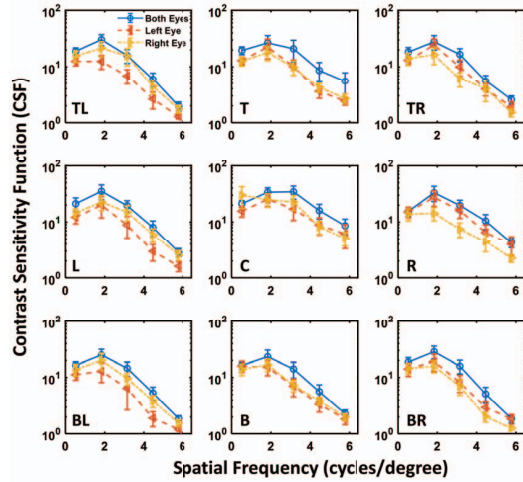


Figure 2: Monocular (red - left eye; yellow - right eye) and binocular (blue - both eyes) contrast sensitivity functions for subjects in group 1 with the appropriate IPD setting ($\Delta\text{IPD} = 0$) measured on the Quest 2 HMD at various locations (TL: top-left, T: top, TR: top-right, L: left, C: center, R: right, BL: bottom-left, B: bottom, and BR: bottom-right).

in two groups on each combination of subject IPD and HMD hardware setting: Group 1: Subject IPD $\approx$ HMD IPD. All 9 subjects participated in the Group 1 experiments by properly adjusting the HMD IPD setting matching their own IPD. Group 2: Subject IPD $>$ HMD IPD. The three subjects with large IPD (mean IPD of 70.3 mm) participated in this set of experiments by setting the HMD IPD to the minimum of 58 mm, i.e., $\Delta\text{IPD} \approx 12$ mm.

Each human observer experiment contains 45 trials (9 target locations $\times$ 5 Gabor spatial frequencies). The subject was instructed to manually adjust the target contrast to the minimum visible contrast using a HMD controller. The reciprocal of the measured threshold contrast at a specified location and spatial frequency was recorded. For each subject, the above experiment was repeated three times binocularly with both eyes open, and monocularly with only left or right eye open by wearing an eye patch on the other eye underneath the headset. We then obtained the contrast sensitivity functions (CSFs) over the spatial frequencies.

3 RESULTS

For human subjects using the appropriate IPD configuration on the HMD (Group 1 subjects), as shown in Fig. 2, the CSF is optimized at the center of display FOV. It has been reported that optical aberration of the HMD lens degrades the contrast and effective resolution at the periphery of VR display FOV [6], resulting in degraded CSFs for targets away from the center location. IPD misalignment, where the IPD of the subject does not match physical lens displacement of the HMD, negatively affects the monocular image quality. Fig. 3 shows that interocular contrast sensitivity variation is substantially pronounced for subjects with larger IPD than that of the HMD (Group 2 subjects, $\Delta\text{IPD} \approx 12$ mm), where the measured CSF favors nasal eye rotation (e.g., right eye rotating towards left). It is also demonstrated that binocular CSF does not always equal to the monocular perception. The difference is particularly enhanced for misaligned IPD between the human subject and HMD optics.

4 CONCLUSION

We develop a WebXR test platform to evaluate monocular and binocular contrast sensitivity. For monocular perception, contrast sensitivity decreases at the periphery of display FOV. For subject with

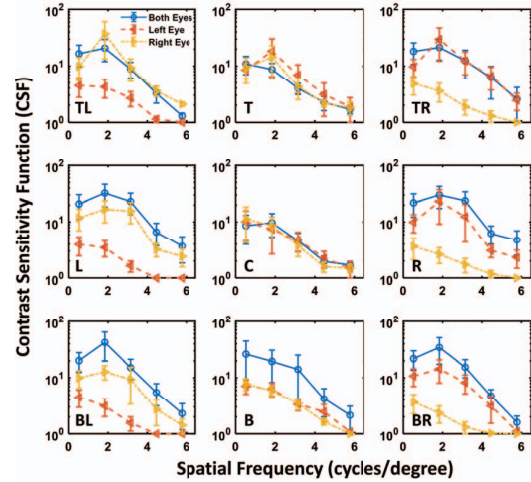


Figure 3: Monocular and binocular contrast sensitivity functions for subjects in group 2 with IPD greater than that of the HMD ($\Delta\text{IPD} = 12$ mm) measured at various locations on the Quest 2 HMD.

larger IPD than the HMD optics, monocular vision favors nasal eye rotation leading to substantial difference in CSFs between left and right eye. Binocular vision is dominated by the superior contrast sensitivity of the two eyes. In future work, a staircase yes-no detection method will be investigated that may potentially minimize the subject input to reduce the random error [5].

ACKNOWLEDGMENTS

Khushi Bhansali was supported by an appointment to the Research Participation Program at the Center for Devices and Radiological Health administrated by the Oak Ridge Institute for Science and Education through an interagency agreement between the U.S. Department of Energy and the U.S. Food and Drug Administration. The mention of commercial products, their sources, or their use in connection with material reported herein is not to be construed as either an actual or implied endorsement of such products by the Department of Health and Human Services.

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